Lab 05 (Thevenin and Norton Equivalent)

Resistors you need: $10\Omega \times 1$; $30\Omega \times 4$; $50\Omega \times 2$; $100\Omega \times 2$; $200\Omega \times 1$

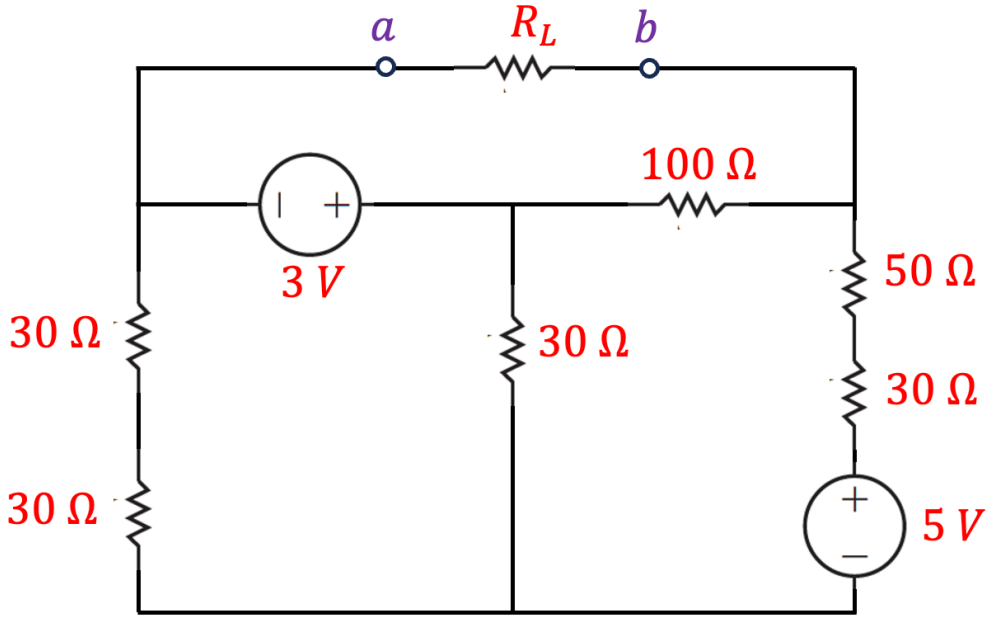


Table 1: Thevenin and Norton Equivalant

	Task 1(calculted)	Task 2(Breadboard)	Task 3(CircuitLab)
v_{oc}			
i_{sc}			
R_{eq}			

Table 2: Voltages across R_L under original circuit

Voltage	$R_L = 10\Omega$	$R_L = 50\Omega$	$R_L = 200\Omega$
Task 2(Breadboard)			
Task 3(CircuitLab)			

Table 3: Voltages across R_L under Thevenin Equivalent

Voltage	$R_L = 10\Omega$	$R_L = 50\Omega$	$R_L = 200\Omega$
Task 1(Calculated)			
Task 2(Breadboard)			
Task 3(CircuitLab)			

Task 1: Calculate the Thevenin and Norton equivalents connected to resistor R_L. Record your results in Table 1 on the first page. Please show your detailed steps. And draw the two equivalents out.

Connect R_L to the Thevenin Equivalent you got. Calculate the voltage across R_L, when $R_L = 10\Omega, 50\Omega, 200\Omega$. Fill the data in Table 3 on the first page.

Task 2: Disconnect R_L.Wire the circuit on your breadboard.
Measure the open-circuit voltage between terminal a and b, v_{oc};
Next, measure the short-circuit current of terminal a and b, i_{sc}.
Finally, turn off all independent sources and measure the equivalent resistor R_{eq} seen between terminal a and b. Record all data in the Table 1 on the first page.
Paste your circuit down.

Now, connect R_L to the original network. Record the voltage across R_L, when $R_L = 10\Omega, 50\Omega, 200\Omega$. Fill the data in Table 2 on the first page.

Last, connect R_L to the Thevenin Equivalent you got in this task. Record the voltage across R_L, when $R_L = 10\Omega, 50\Omega, 200\Omega$. Fill the data in Table 3 on the first page.

Task 3: Disconnect R_L. Build the original circuit in CircuitLab, record v_{oc} and i_{sc} , and then calculate $R_{eq} = v_{oc}/i_{sc}$ Record all data in the Table 1 on the first page.
(Hint: The current in a wire cannot be measured directly. Measure the currents in the connected branches and apply KCL to compute the current of interest.)
Paste your circuit down.

Now, connect R_L to the original network. Record the voltage across R_L, when $R_L = 10\Omega, 50\Omega, 200\Omega$. Fill the data in Table 2 on the first page.

Last, connect R_L to the Thevenin Equivalent you got in this task. Record the voltage across R_L, when $R_L = 10\Omega, 50\Omega, 200\Omega$. Fill the data in Table 3 on the first page.